Validation and Failure Testing

Nine failure tests run during continuous pings, with procedures, results and screenshots.

Blake Otto · Catalyst Switch Lab — Small Office Network Build
github.com/blakeotto57/Catalyst-Switch-Lab-Small-Office-Network-Build

Revised October 10, 2026. The switch login name is blacked out in screenshots.

Failure tests

Each test ran during a continuous ping from the test laptop on SW2. The laptop and the server VM shared one network adapter, so most tests pinged SW3's management address (10.10.99.3) instead of the server; that traffic takes the same path through R1.

Result: 8 of 9 tests passed, and test 8 passed in part.

#	Test	Result
1	One link in SW2's EtherChannel fails	Pass — Po1 stayed up on the remaining link; the captured ping window shows no timeouts
2	SW2 loses both core uplinks	Pass — traffic resumed over the cross-link after about 36 s (9 timed-out pings at Windows' 4 s default timeout); SW3 runs 802.1D, so its port has to pass through listening and learning
3	SW2's uplinks are restored	Pass — Po1 re-formed with both links and became SW2's root port again; SW3 Fa0/4 returned to blocking
4	Cross-link role check	Pass — SW3 Fa0/4 blocking, SW2 Gi0/4 designated forwarding
5	BPDU received on an edge port	Pass — %SPANTREE-2-BLOCK_BPDUGUARD and %PM-4-ERR_DISABLE logged; test port err-disabled
6	Native VLAN mismatch	Pass — %CDP-4-NATIVE_VLAN_MISMATCH logged; an earlier accidental mismatch (VLAN 1 vs 900 after SW3 reloaded) was also blocked by spanning tree's PVID-inconsistency check on both switches
7	Root bridge change	Pass — SW2 became root for VLAN 10, the blocked port moved from SW3 Fa0/4 to SW3's uplink (cross-link cost 19 < 12 + 12 via R1), and both returned when the priority was restored
8	Lab device attempts to reach a home device	Partial — blocked as required (ACL line 10 hit counter and %SEC-6-IPACCESSLOGDP) while lab-internal traffic worked; the "internet still works" half couldn't be tested (see requirement 10)
9	Second device on a secured port (optional)	Pass — %PORT_SECURITY-2-PSECURE_VIOLATION named the second MAC; the port went to Secure-shutdown and recovered with shutdown / no shutdown

Test 1: One link in SW2's EtherChannel fails

What it proves: EtherChannel redundancy

Pass criteria: Traffic continues over the remaining link and the bundle stays up

Result: Pass — Po1 stayed up on the remaining link; the captured ping window shows no timeouts

Procedure and evidence

Before the test, SW2's EtherChannel summary:

```
SW2#sh ether sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

and R1's:

```
R1#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

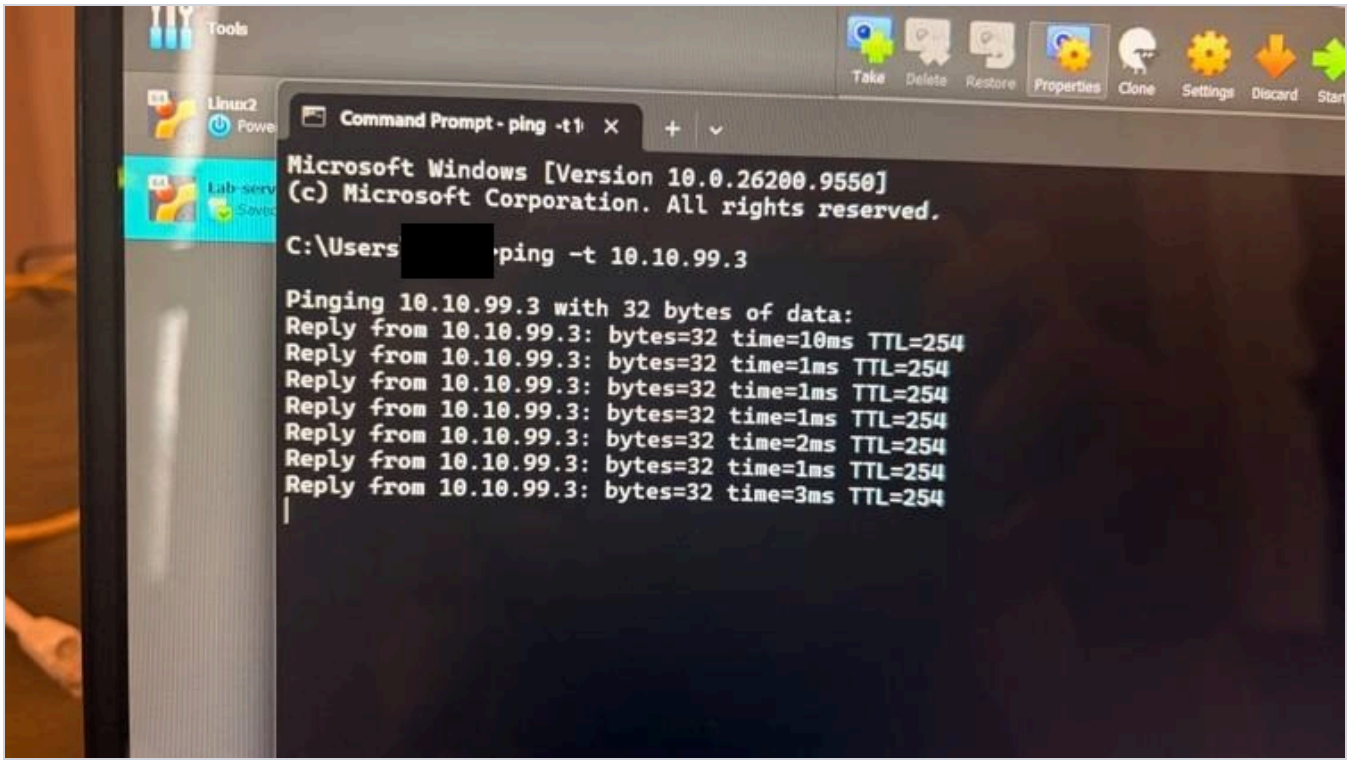
        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

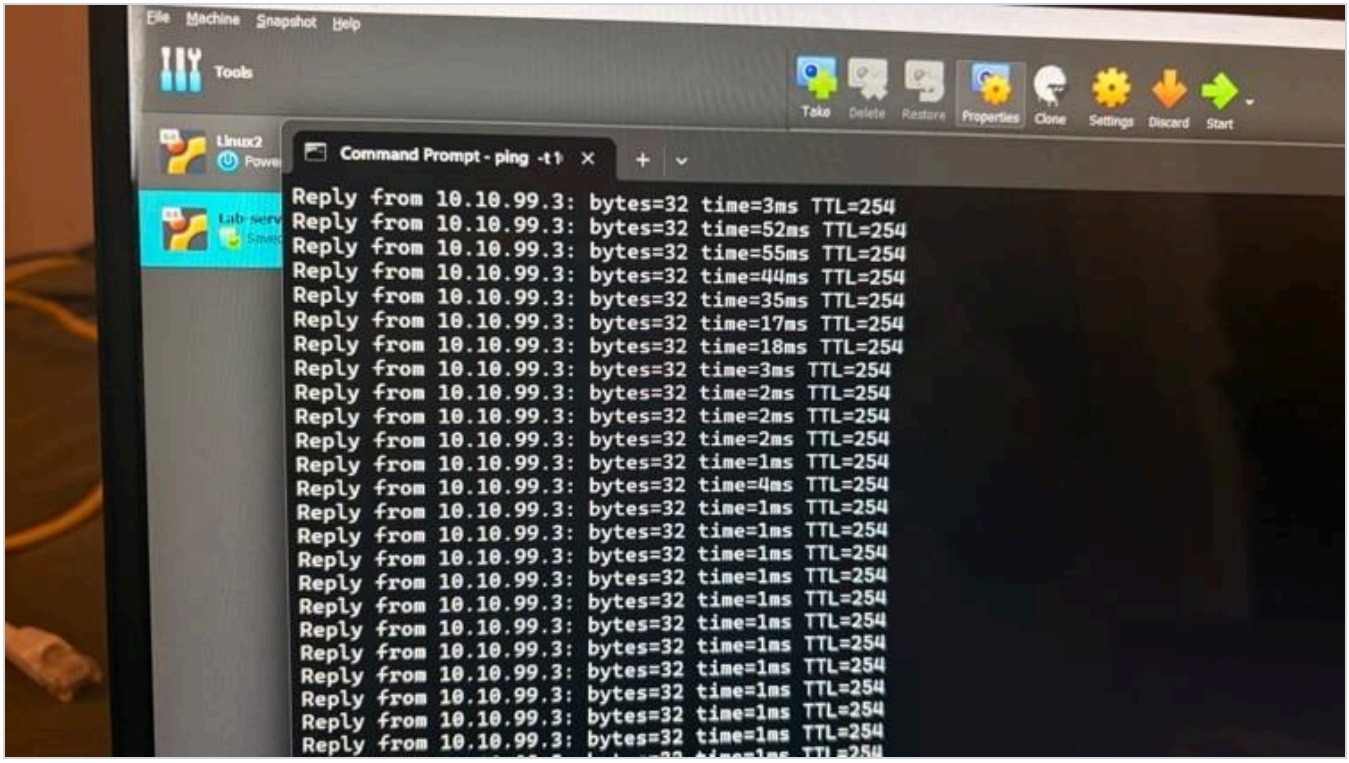
Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------
1      Po1(SU)         LACP      Fa0/1(P)    Fa0/2(P)
2      Po2(SU)          -        Fa0/3(P)    Fa0/4(P)

R1#
```

On the test laptop connected to SW2 Gi0/3, I started `ping -t 10.10.99.3` (SW3's management IP):



I then unplugged Gi0/1 on SW2. Replies kept coming; latency rose briefly around the unplug, and no request timed out in the captured window:



SW2's summary with one link down: Po1 is still up (SU) on Gi0/2, and Gi0/1 shows D (down):

```
SW2#sh eth su
*Mar  1 03:40:30.948: %SYS-5-CONFIG_I: Configured from console by [redacted] on console
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SU)         LACP      Gi0/1(D)    Gi0/2(P)

SW2#
```

After I plugged Gi0/1 back in, both links were bundled again:

```
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

The capture has no Ctrl+C summary, so the exact loss count isn't recorded.

Test 2: SW2 loses both core uplinks

What it proves: Backup path through the cross-link

Pass criteria: Spanning tree unblocks the cross-link and traffic reaches the core through SW3; convergence time is recorded

Result: Pass — traffic resumed over the cross-link after about 36 s (9 timed-out pings at Windows' 4 s default timeout); SW3 runs 802.1D, so its port has to pass through listening and learning

Procedure and evidence

Before the test, SW2's root port is Po1 and the cross-link Gi0/4 is designated forwarding:

```
SW2#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

SW3 is root for VLAN 1, which only SW3's trunks carry (R1 and SW2 don't allow it), so that instance doesn't affect the lab VLANs:

```
SW3#sh span

Spanning tree 1 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6200
  Configured hello time 2, max age 20, forward delay 15
  We are the root of the spanning tree
  Topology change flag not set, detected flag not set, changes 2
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 1, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 1 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 32768, address 0007.0ecc.6200
   Designated bridge has priority 32768, address 0007.0ecc.6200
   Designated port is 2, path cost 0
   Timers: message age 0, forward delay 0, hold 0
   BPDU: sent 6738, received 0

Interface Fa0/4 (port 16) in Spanning tree 1 is FORWARDING
   Port path cost 19, Port priority 128
   Designated root has priority 32768, address 0007.0ecc.6200
   Designated bridge has priority 32768, address 0007.0ecc.6200
   Designated port is 16, path cost 0
   Timers: message age 0, forward delay 0, hold 0
   BPDU: sent 3600, received 0

 --More--
```

For VLAN 10, SW3's root port is its uplink and the cross-link Fa0/4 is blocking:

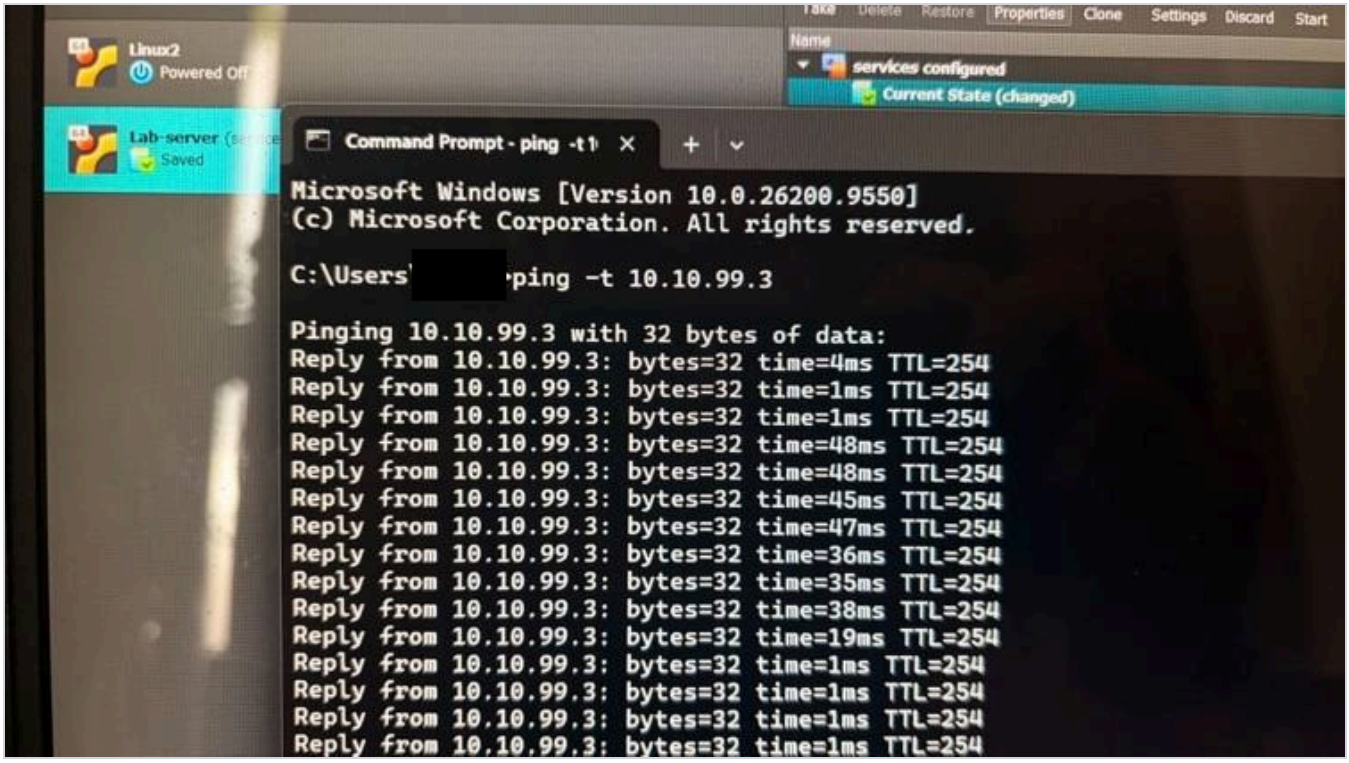
```
SW3#show spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 7
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

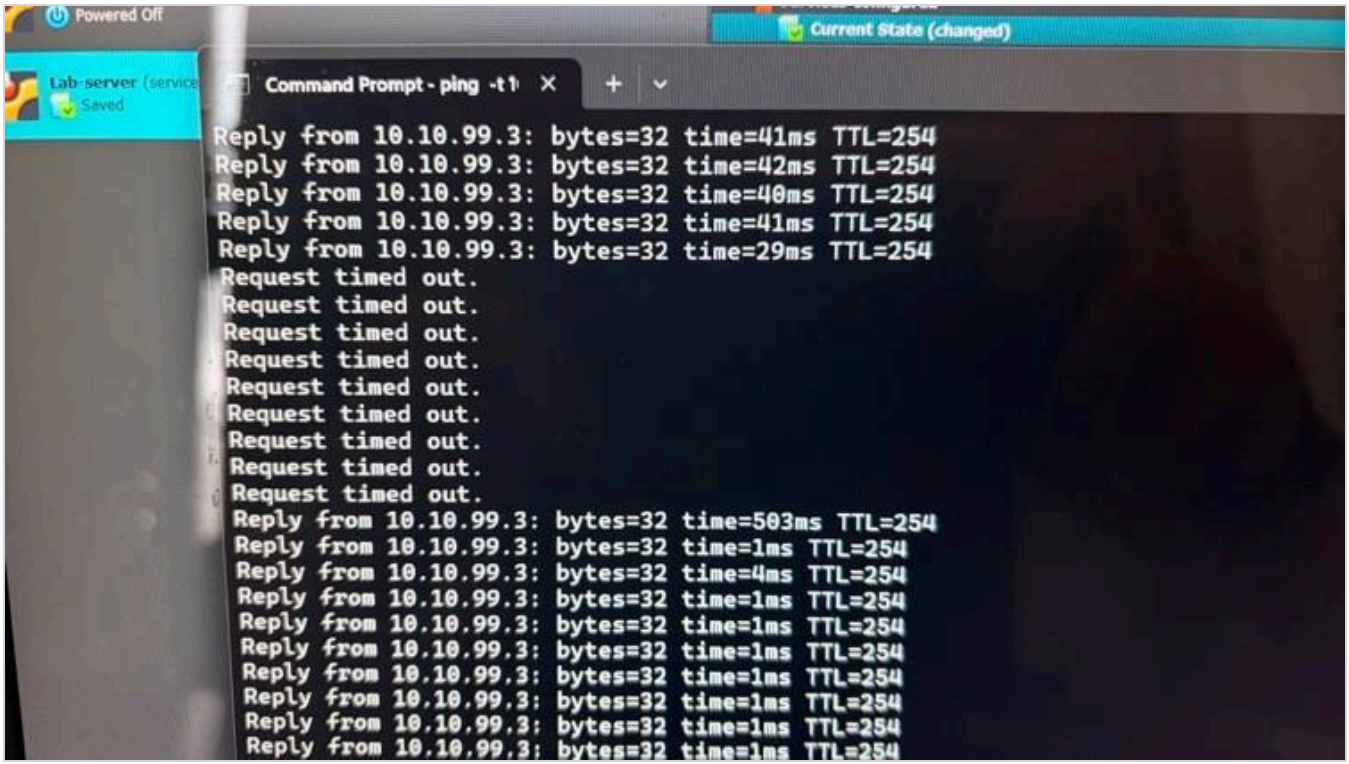
Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 5, received 6800

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 89, received 3572
SW3#
```

I pinged 10.10.99.3 from the laptop on SW2, then unplugged both EtherChannel links (Gi0/1–2) on SW2. Before the unplug:



After the unplug, 9 requests timed out, about 36 seconds at Windows' default 4-second timeout. Then replies resumed over the cross-link:



SW3's Fa0/4 is now forwarding and designated, carrying SW2's traffic to the core through SW3. I captured the blocking state before and the forwarding state after, but not the listening and learning states in between (802.1D spends 15 seconds in each by default):

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 8
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 8, received 7292

Interface Fa0/4 (port 16) in Spanning tree 10 is FORWARDING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 32768, address 0007.0ecc.6201
   Designated port is 16, path cost 12
   Timers: message age 0, forward delay 0, hold 0
   BPDU: sent 150, received 4006
SW3#
```

I then plugged both links back in; test 3 shows the recovery.

Test 3: SW2's uplinks are restored

What it proves: Recovery to the normal path

Pass criteria: Traffic returns to SW2's own uplinks and the cross-link blocks again

Result: Pass — Po1 re-formed with both links and became SW2's root port again; SW3 Fa0/4 returned to blocking

Procedure and evidence

After I restored SW2's uplinks, the EtherChannel re-formed with both links bundled:

```
SW2#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

SW2 uses Po1 as its root port again, and the cross-link Gi0/4 is designated forwarding:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

SW3's Fa0/4 is back to blocking:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 9
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 9, received 7387

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
 --More--
```

I didn't capture a ping during the restore.

Test 4: Cross-link role check

What it proves: Spanning tree design

Pass criteria: The cross-link port on SW3 is blocking while SW2's side is forwarding

Result: Pass — SW3 Fa0/4 blocking, SW2 Gi0/4 designated forwarding

Procedure and evidence

These captures were taken before test 2, with the network in its normal state. SW2's side of the cross-link (Gi0/4) is designated forwarding:

```
SW2#sh span

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p
```

and SW3's side (Fa0/4) is blocking:

```
SW3#show spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 7
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 5, received 6800

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 89, received 3572
SW3#
```

Test 3's captures show the same roles after SW2's uplinks were restored.

Test 5: BPDU received on an edge port

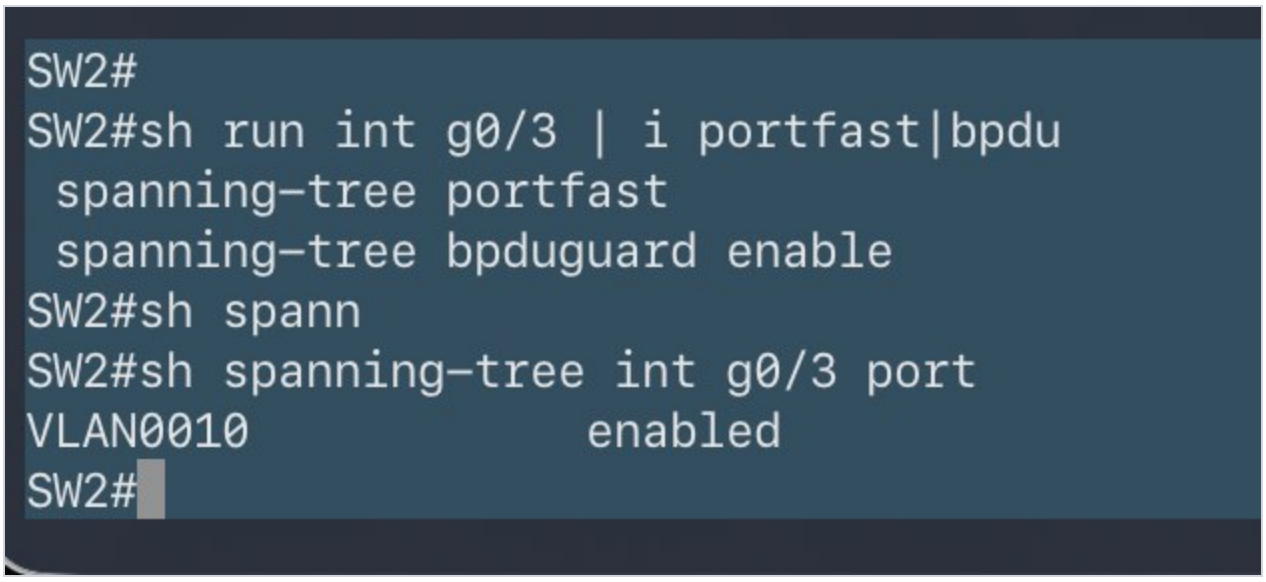
What it proves: Loop protection

Pass criteria: The port is error-disabled and the event is logged

Result: Pass — %SPANTREE-2-BLOCK_BPDUGUARD and %PM-4-ERR_DISABLE logged; test port err-disabled

Procedure and evidence

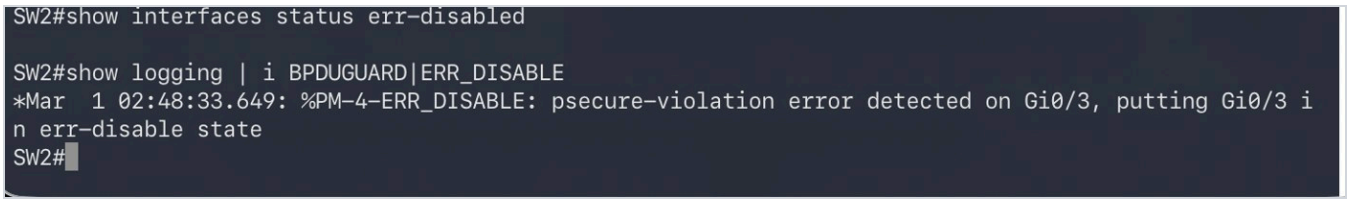
Before the test, SW2's user port Gi0/3 has PortFast and BPDU Guard:



I configured an unused port on SW2 as an edge port for the test:

```
int g0/5
 desc BPDU-GUARD-TEST
 sw mode acc
 sw acc vlan 10
 spanning-tree portfast
 spanning-tree bpduguard en
 no shut
```

With the laptop on SW2 Gi0/3, I connected a cable from SW2 Gi0/5 to SW3 Fa0/5 and started `ping -t -w 1000 10.10.99.3`. The pings succeeded. Then on SW2 I ran `show interfaces status err-disabled` and `show logging | i BPDUGUARD|ERR_DISABLE`:



Nothing showed for Gi0/5 yet (the only entry is an earlier port-security event on Gi0/3), so I checked the port:

```
SW2#show interfaces status | i Gi0/5
Gi0/5     BPDU-GUARD-TEST    notconnect   10           auto   auto 10/100/1000BaseTX
Gi0/50                       disabled     999          auto   auto Not Present
Gi0/51                       disabled     999          auto   auto Not Present
Gi0/52                       disabled     999          auto   auto Not Present
SW2#show run interface g0/5
Building configuration...

Current configuration : 172 bytes
!
interface GigabitEthernet0/5
 description BPDU-GUARD-TEST
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
end
```

SW3's Fa0/5 was still administratively down. After I ran `no shutdown` on it, SW3 sent BPDUs into Gi0/5, and SW2 err-disabled the port and logged it:

```
SW2#show interfaces status err-disabled

Port      Name               Status       Reason               Err-disabled Vlans
Gi0/5     BPDU-GUARD-TEST    err-disabled bpduguard
SW2#show logging | i BPDUGUARD|ERR_DISABLE
*Mar  1 04:27:42.707: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi0/5 with BPDU Guard enab
led. Disabling port.
*Mar  1 04:27:42.707: %PM-4-ERR_DISABLE: bpduguard error detected on Gi0/5, putting Gi0/5 in err-di
sable state
SW2#
```

Afterward I moved Gi0/5 back to VLAN 999 and shut it down.

Test 6: Native VLAN mismatch

What it proves: Trunk misconfiguration detection

Pass criteria: The mismatch is detected and logged by the switches

Result: Pass — %CDP-4-NATIVE_VLAN_MISMATCH logged; an earlier accidental mismatch (VLAN 1 vs 900 after SW3 reloaded) was also blocked by spanning tree's PVID-inconsistency check on both switches

Procedure and evidence

Before the test, SW2 sees SW3 on Gi0/4 through CDP, and the trunk's native VLAN is 900:

```
SW2#sh cdp neigh g0/4
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
SW3.blakeotto.com
                 Gig 0/4           162           T S      WS-C3548- Fas 0/4
SW2#h int g0/4 sw | i Native
help int g0/4 sw | i Native
     ^
% Invalid input detected at '^' marker.

SW2#show int g0/4 sw | i Native
Trunking Native Mode VLAN: 900 (NATIVE)
Administrative Native VLAN tagging: enabled
Administrative private-vlan trunk Native VLAN tagging: enabled
SW2#
```

With `ping -t -w 1000 10.10.99.3` running on the laptop, I created the mismatch on SW2:

```
int g0/4
 sw trunk native vlan 99
 end
```

Then I waited for CDP and checked the log with `show logging | i NATIVE|PVID|INCONSISTENT`. CDP detected the mismatch:

```
SW2#show logging | i NATIVE|PVID|INCONSISTENT
*Mar  1 04:34:19.161: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthern
et0/4 (99), with SW3.blakeotto.com FastEthernet0/4 (900).
SW2#show logging | i NATIVE|PVID|INCONSISTENT
```

Then I restored the native VLAN to 900:

```
SW2(config)#int g0/4
SW2(config-if)#sw trunk nat
SW2(config-if)#sw trunk native vla
SW2(config-if)#sw trunk native vlan 900
SW2(config-if)#end
SW2#
*Mar  1 04:36:14.739: %SYS-5-CONFIG_I: Configured from console by [redacted] on console
SW2#show int g0/4 sw | i Native
Trunking Native Mode VLAN: 900 (NATIVE)
Administrative Native VLAN tagging: enabled
Administrative private-vlan trunk Native VLAN tagging: enabled
SW2#
```

Earlier in the build, SW3 reloaded with an unsaved config, and its Fa0/4 came back as an access port in VLAN 1. Both switches caught that mismatch too. From the terminal logs I saved at the time (the device clocks weren't synchronized):

```
SW2
*Mar  1 04:59:56.138: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
GigabitEthernet0/4 (900), with SW3.blakeotto.com FastEthernet0/4 (1).
*Mar  1 05:04:09.927: %SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan id
1 on GigabitEthernet0/4 VLAN900.
*Mar  1 05:04:09.927: %SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/4 on VLAN0900.
Inconsistent local vlan.
*Mar  1 05:04:40.940: %SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/4 on
VLAN0900. Port consistency restored.

SW3
*Mar  1 00:13:30.944: %SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan id
900 on FastEthernet0/4 on vlan 1.
*Mar  1 00:13:30.947: %SPANTREE-2-BLOCK_PVID_PEER: Blocking FastEthernet0/4 on vlan 900.
Inconsistent peer vlan.
*Mar  1 00:13:30.950: %SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/4 on vlan 1.
Inconsistent local vlan.
*Mar  1 00:14:04.213: %SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/4 on vlan
900. Port consistency restored.
```

Test 7: Root bridge change

What it proves: Spanning tree control

Pass criteria: The root moves as configured, the blocked port changes accordingly, and both return when priorities are restored

Result: Pass — SW2 became root for VLAN 10, the blocked port moved from SW3 Fa0/4 to SW3's uplink (cross-link cost 19 < 12 + 12 via R1), and both returned when the priority was restored

Procedure and evidence

Before the test, VLAN 10 on SW3 (root port Fa0/2 toward R1, cross-link Fa0/4 blocking):

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 2
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 1, received 413

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 3, forward delay 0, hold 0
   BPDU: sent 1, received 34
SW3#
```

and on SW2:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

On SW2 I lowered its VLAN 10 priority with `spanning-tree vlan 10 priority 20480`, and SW2 became root for VLAN 10:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    20490
             Address     0019.e885.9000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    20490  (priority 20480 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Desg FWD 12        128.56   P2p


SW2#
```

SW3 now reaches the root over the cross-link: Fa0/4 is forwarding as its root port (cost 19), and its uplink Fa0/2 is blocking, because 19 is less than the 12 + 12 path through R1:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 20490, address 0019.e885.9000
  Root port is 16, cost of root path is 19
  Topology change flag not set, detected flag not set, changes 3
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is BLOCKING
   Port path cost 12, Port priority 128
   Designated root has priority 20490, address 0019.e885.9000
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 12
   Timers: message age 3, forward delay 0, hold 0
   BPDU: sent 2, received 525

Interface Fa0/4 (port 16) in Spanning tree 10 is FORWARDING
   Port path cost 19, Port priority 128
   Designated root has priority 20490, address 0019.e885.9000
   Designated bridge has priority 20490, address 0019.e885.9000
   Designated port is 4, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 2, received 146
SW3#
```

R1 is no longer root; its root port is Po1, toward SW2:

```
R1#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    20490
             Address     0019.e885.9000
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0016.9d94.0980
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Root FWD 12        128.56   P2p
Po2                 Desg FWD 12        128.64   P2p Peer(STP)


R1#
```

To restore, I set SW2 back to `spanning-tree vlan 10 priority 28672`. SW2 is no longer root:

```
SW2#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

SW3 and R1 are back to their original roles:

```
SW3#sh spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 4
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 1, forward delay 0, hold 0
   BPDU: sent 5, received 648

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 3, received 267
SW3#
```

```
R1#sh span
R1#sh spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24586  (priority 24576 sys-id-ext 10)
             Address     0016.9d94.0980
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Po1                 Desg FWD 12        128.56   P2p
Po2                 Desg FWD 12        128.64   P2p Peer(STP)


R1#
```

Test 8: Lab device attempts to reach a home device

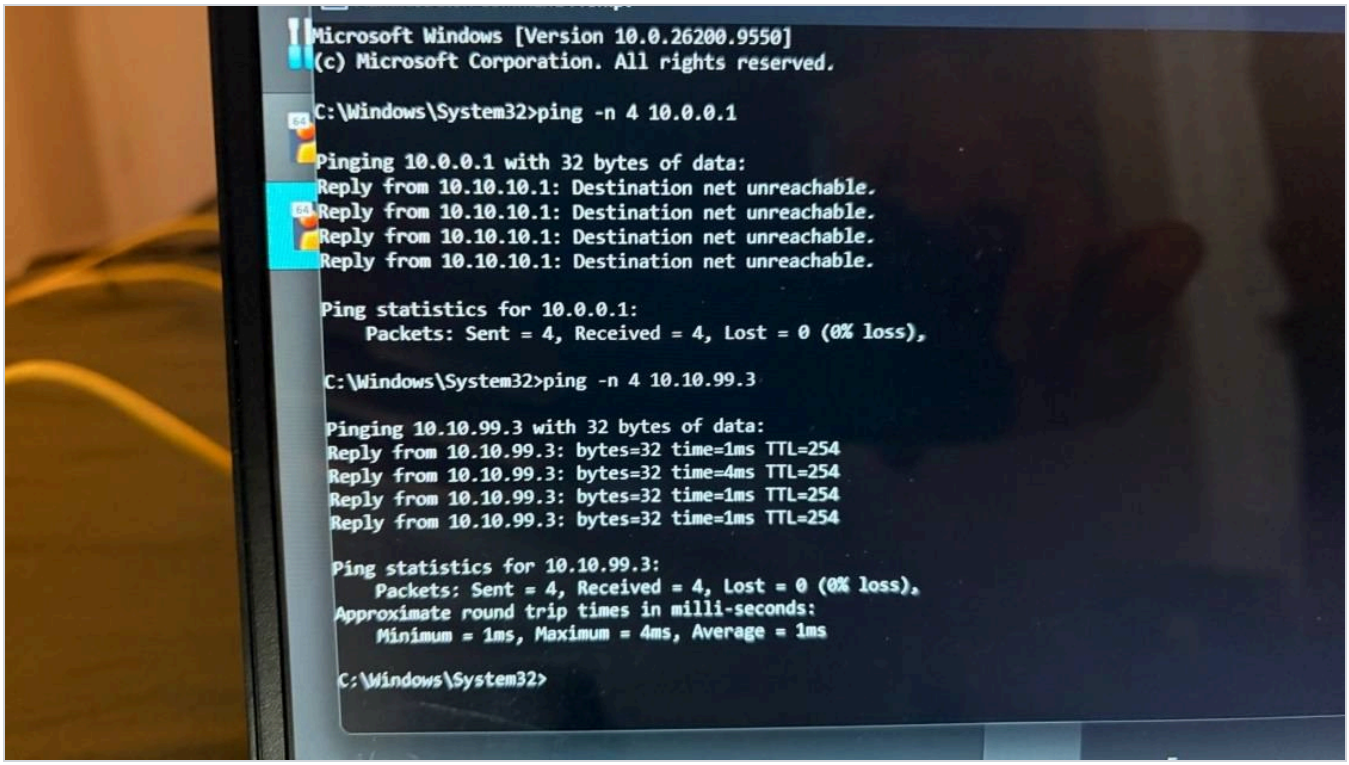
What it proves: Access list enforcement

Pass criteria: The connection is blocked while internet access still works

Result: Partial — blocked as required (ACL line 10 hit counter and %SEC-6-IPACCESSLOGDP) while lab-internal traffic worked; the "internet still works" half couldn't be tested (see requirement 10)

Procedure and evidence

On R1 I cleared the ACL counters with `clear access-list counters LAB_BLOCK_HOME`. Then, from the laptop on SW2, I pinged the home router (should fail) and SW3's management IP (should succeed):



R1's ACL counted the denied packets on line 10 (VLAN 10) and logged them:

```
R1#show access-lists LAB_BLOCK_HOME
Extended IP access list LAB_BLOCK_HOME
    10 deny ip 10.10.10.0 0.0.0.255 10.0.0.0 0.0.0.255 log (8 matches)
    20 deny ip 10.10.20.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    30 deny ip 10.10.99.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    40 permit ip any any
R1#show logging | i IPACCESSLOG
*Mar  1 13:49:10.264: %SEC-6-IPACCESSLOGDP: list LAB_BLOCK_HOME denied icmp 10.10.10.21 -> 10.0.0.1 (0
/0), 1 packet
R1#
```

The block works, and lab-internal traffic is unaffected. The other half of the pass criteria, "internet access still works," couldn't be tested, because lab hosts have no path to the internet (requirement 10). So this test passed only in part.

Test 9: Second device on a secured port (optional)

What it proves: Port security

Pass criteria: The violation is detected and logged

Result: Pass — %PORT_SECURITY-2-PSECURE_VIOLATION named the second MAC; the port went to Secure-shutdown and recovered with shutdown / no shutdown

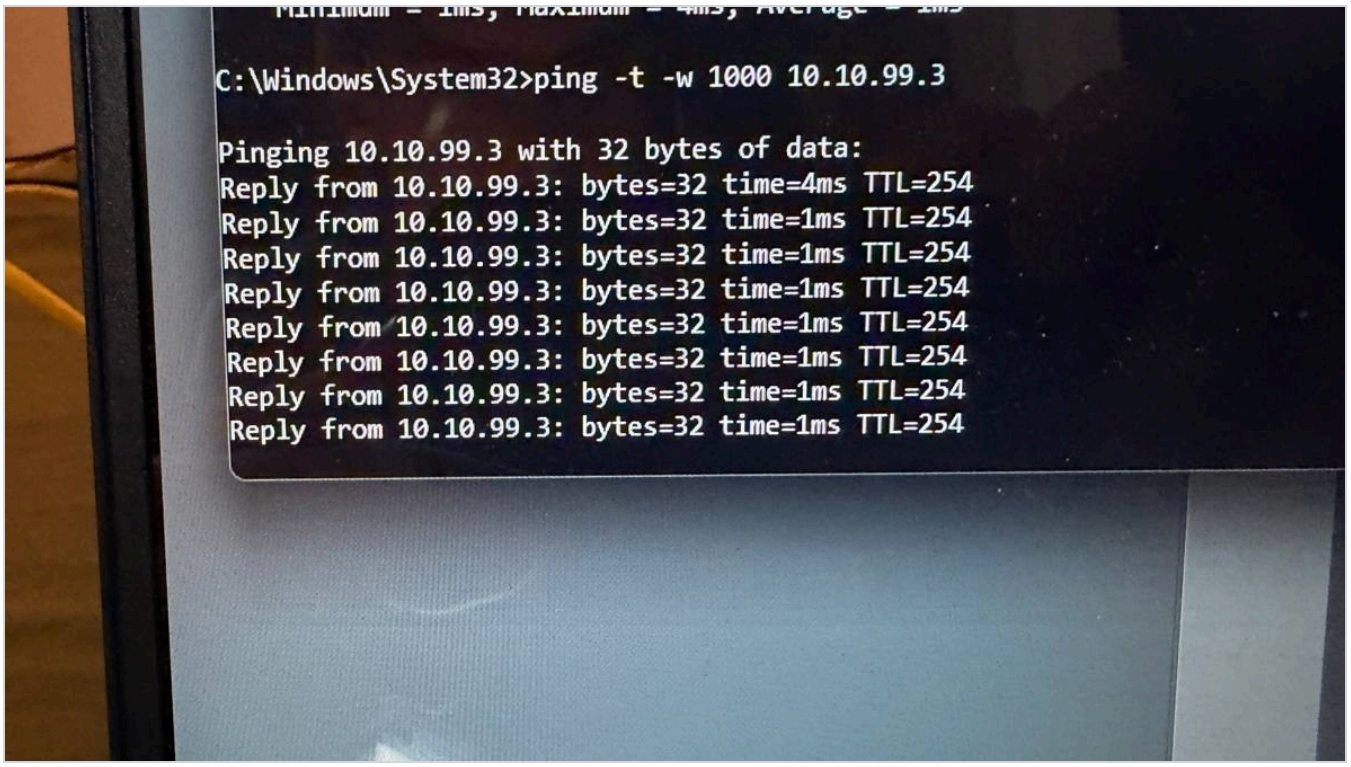
Procedure and evidence

Before the test, port security on SW2 Gi0/3 (the laptop's port) allows one MAC, learned as sticky:

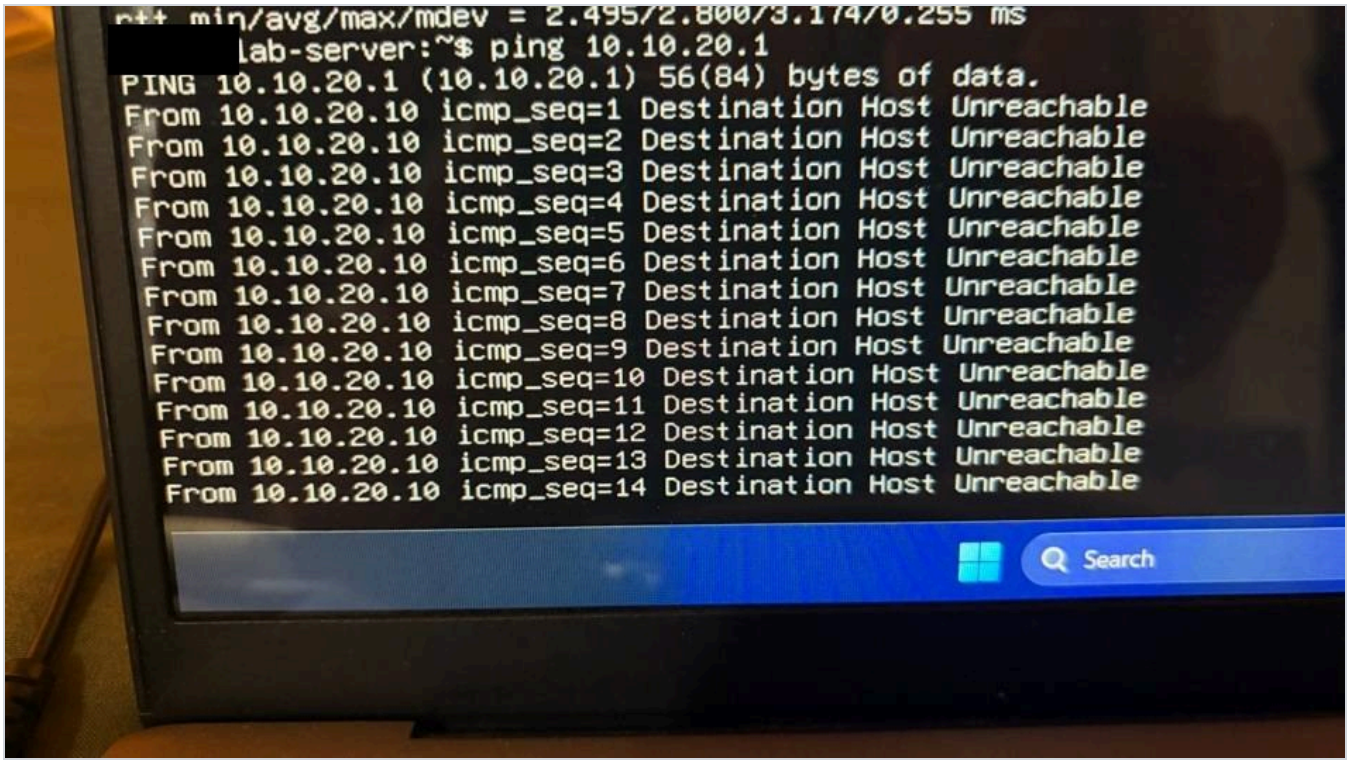
```
SW2#sh port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0800.2749.db1d:10
Security Violation Count   : 0

SW2#
```

I pinged 10.10.99.3 from the Windows laptop, then pinged from the Ubuntu VM, which uses the same USB adapter, so its MAC appeared as a second address on Gi0/3. On the laptop:



On SRV1, the ping to its gateway (10.10.20.1) fails. That alone doesn't prove much, because SRV1 is addressed for VLAN 20 and Gi0/3 is a VLAN 10 port:



The switch side is the proof: SW2 logged %PORT_SECURITY-2-PSECURE_VIOLATION for the VM's MAC (0800.2749.db1d), Gi0/3 is in Secure-shutdown, and the violation count went up:

```
SW2#show mac address-table interface g0/3
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
SW2#sh port
SW2#sh port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0800.2749.db1d:10
Security Violation Count   : 1

SW2#show logging | i PSECURE|ERR_DISABLE
*Mar  1 05:26:30.721: %PM-4-ERR_DISABLE: psecure-violation error detected on Gi0/3, putting Gi0/3 in e
rr-disable state
*Mar  1 05:26:30.729: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC a
ddress 0800.2749.db1d on port GigabitEthernet0/3.
SW2#
```

To recover, I shut down the VM and ran `shutdown` and `no shutdown` on Gi0/3. The port is back up:

```
SW2(config)#int g0/3
SW2(config-if)#shut
SW2(config-if)#no shut
SW2(config-if)#
SW2(config-if)#end
SW2#sh por
*Mar  1 05:34:40.112: %SYS-5-CONFIG_I: Configured from console by [redacted] on console
*Mar  1 05:34:40.548: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed stat
e to up
SW2#sh port
SW2#sh port-security
*Mar  1 05:34:41.873: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
SW2#sh port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0800.2749.db1d:10
Security Violation Count   : 0

SW2#
```